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Multi-scale triple-attention network for pixelwise crack segmentation

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ABSTRACT

Currently, intelligent crack detection is of great value for the maintenance of infrastructure, of which the most significant kind in China is roads. For pavement defects, the pavement can be repaired and maintained in a timely manner with an accurate defect detection task, which significantly reduces the occurrence of hazards. However, the detection of pavement defects remains a great challenge owing to many difficulties, for example, complex backgrounds, microdefects, various defect shapes and sizes, class imbalance issues, etc. Recently, deep learning has demonstrated its superior performance on pixelwise image segmentation, but some issues still exist on demanding pixelwise image segmentation, for instance, limited receptive field, insufficiency processing of local features, information loss issue generated by pooling operations, etc. Based on all of the above issues, a multiscale triple-attention network, named MST-Net, is proposed for end-to-end pixelwise crack detection. First, a multiscale input strategy is applied to the proposed segmentation network to capture more context information. Meanwhile, it can capably reduce the effect of the information loss issue generated by pooling operations. Second, to realize effective feature representation of local features, an additive attention fusion (AAF) block is proposed to guide feature learning to capture both global and local contexts. In addition, faced with the crack detection task with class imbalance issues, a triple attention (TA) block is proposed to detect spatial, channel and pixel attention information to suppress the background and useless information, which is conducive to the characterization of microcracks. Finally, aiming at the limited receptive field, a multiscale feature aggregation unit is proposed for feature fusion to increase the detection ability of multiscale defects. To better guide network training, a deep supervision mechanism is also introduced to speed up the convergence of the proposed segmentation model and improve the performance of defect segmentation. The related evaluation and detection experiments are carried out on three public datasets on crack segmentation, and the comparison experiments with the mainstream segmentation models show that the proposed segmentation network achieves excellent performance on pixelwise crack detection.

1. Introduction

With the continuous development of the modern economy, the construction of infrastructure has also risen to a new level. Among the many kinds of infrastructure, the most common one is the construction of roads, in which the government has invested many financial and material resources, thus comprehensively driving the development and progress of the economy. However, after a long period of sun and rain and the crushing of passing vehicles, the road surface is prone to various problems, the most common of which is cracks. If not repaired by professionals in a timely manner, the lighter damage to the road surface will be exacerbated, affecting the quality of passage, and the heavier damage will cause the road surface to collapse, resulting in accidents and endangering the lives and health of working people. Therefore, to prolong the service life of roads and prevent accidents, accurate and effective pavement defect detection methods are necessary. The traditional crack detection method has been always based on manual detection, and the detection process is laborious and time-consuming and cannot be used in the production process of modern industry. Moreover, the inspection process is based on the subjective consciousness of the inspection workers. When the inspection worker is mentally fatigued, there is a tendency to make false detections or miss the cracks, which has a great impact on the inspection accuracy. These cracks are always faced with challenging factors, such as complex backgrounds, microdefects, various defect shapes and sizes, and class imbalance issues, which present some difficulties for accurate crack detection. Therefore, until now, accurate crack detection has still been challenging for some researchers and industrial product producers [1].

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At present, after years of research and development, various testing methods have been adopted for nondestructive crack detection [2], and the above nondestructive crack detection methods can be roughly divided into two categories: model-based methods [3,4] and deep learning-based methods. In the early days of focusing on pavement crack segmentation, model-based segmentation methods first appeared in the public's view and achieved remarkable results. Currently, the most prevalent and widely used model-based methods include thresholding-based [5] and edge detection-based [6] methods, which are the most representative detection methods based on image processing. Henrique et al. proposed a simple unsupervised crack detection system [7] for reducing the variance caused by uneven pixels. Here, a morphological filter was used to preprocess the defect images, and the dynamic threshold method was used to identify the crack pixels. To detect weak edges that have little change with the background gray, Zhao et al. proposed an improved Canny operator for edge detection [8]. The wavelet transform was used to strengthen the weak edge of the input image, and the genetic algorithm was used for the Canny algorithm to obtain a suitable threshold. Li et al. proposed a crack defect recognition method based on the double-threshold segmentation method [9]. The Otsu algorithm was used to remove the redundant road markings, and the adaptive threshold iterative segmentation algorithm was proposed to obtain the crack image. Meanwhile, due to a large amount of noise in the image, which could affect the accuracy of segmentation, the contours of objects were obtained by morphological denoising. The above model-based methods always segment cracks from defect images based on specific image characteristics, such as edge, texture and color, which are greatly affected by complex factors such as image noise, shadow, complex backgrounds, uneven illumination and so on, leading to poor accuracy and robustness in pixelwise defect detection.

In recent years, deep learning-based methods have developed rapidly and have demonstrated excellent performance in various computer vision tasks, such as image classification [10], object detection [11], and image segmentation [12–14]. Based on their excellent performance, a wide range of applications have also been achieved in the task of detecting road surface cracks and defects with good performance in terms of detection efficiency and quality [15–17]. To capture weak features of pavement microcracks and eliminate background interference, Liu et al. [18] proposed an end-to-end feature fusion codec network with two new modules to improve crack detection accuracy and introduced a feature fusion module and a shape semantic prior module in the segmentation network. To address the problems of detail loss and incomplete crack extraction during pavement crack detection, Qu et al. [19] proposed a deep network model containing a hierarchical feature fusion based on attention structures and a residual network as the baseline network to automatically and accurately detect pavement cracks with uneven intensity distribution. To distinguish the similarity between pavement defects and the background and avoid the undersegmentation issue, Gao et al. [20] proposed a pixel-level defect segmentation network, fusing the low, medium and high levels of the proposed baseline ResNet50 network to more effectively represent the detailed information of the defects. To address the problems of insufficient datasets and long-narrow shapes in pavement crack segmentation, Chen et al. [21] proposed a lightweight hybrid convolutional neural network model to achieve road crack segmentation with fewer parameters. Although the network model in the above process has improved the accuracy of pavement crack segmentation to a certain extent and plays a positive role in the maintenance and detection of pavements, there are still a number of problems, such as insufficient detection of detail features, influenced by the background light, etc. Compared with the end-to-end crack detection framework proposed above, a network model based on the encoder–decoder structure is also proposed to segment the image. The U-Net network is a superior typical encoder–decoder segmentation network with a skip connection structure between the encoder and decoder to effectively aggregate different feature information from the encoder and decoder, which has been widely applied in a variety of segmentation applications, such as pavement crack segmentation.

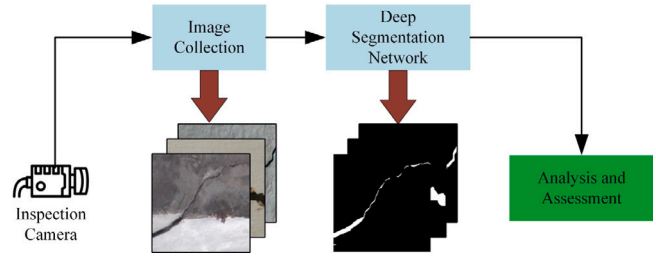

Fig. 1. Network framework of the proposed defect detection.

In response to the many problems of pavement crack detection, researchers have taken advantage of the U-Net network structure and proposed a number of variants of the U-Net network to further improve the superior performance of pavement crack detection. In response to the complexity of the concrete pavement texture background and the multiple forms of cracks, Qu et al. [22] proposed an improved U-Net model with a residual attention mechanism in the encoder part and a pyramid feature fusion structure in the decoder stage to effectively segment pavement cracks, but the segmentation network contained a relatively large number of convolutional and pooling layers, which could easily cause information loss issues. To address the problems of pavement crack interruptions caused by conventional segmentation networks and the emergence of discrete background noise, Han et al. [23] proposed a deformation U-Net network to achieve pixel-level semantic segmentation of pavement cracks, but the network was insensitive to the class imbalance problem of pavement cracks. To overcome the problem of low accuracy and poor stability in crack detection, Qiao et al. [24] proposed an improved U-Net convolutional neural network with an improved initial module and an atrous spatial pyramidal pooling module, which effectively improved the accuracy and increased the robustness. However, the network model was not particularly sensitive to the boundary information of cracks, and the detected crack boundaries were not clear. To focus on pixel areas of cracks, Augustauskas et al. [25] proposed an improved attention U-Net network with an atrous spatial pyramid pooling module for pixel-level pavement-defect segmentation. In summary, the previously proposed U-Net variant-based networks achieve superior performance in pavement crack segmentation and have a positive impact on the maintenance of pavement. However, from the above analysis, the above networks still have some problems with high-precision crack segmentation, such as a limited receptive field, insufficient processing of local features, and information loss issues generated by pooling operations.

From the above analysis of crack detection, this paper proposes a multiscale triple-attention network, named MST-Net, for an end-to-end and accurate pixelwise crack detection network to serve quality control and maintenance decisions. Fig. 1 shows the diagram of the proposed crack detection scheme. Combined with three public crack defect datasets, it has demonstrated an excellent performance on pixelwise crack detection in view of the interference of complex backgrounds, especially for accurate detection of microcracks. The main contributions of this paper are as follows.

(1) An end-to-end defect segmentation network is proposed for crack detection to locate pixel-level cracks automatically and accurately, as illustrated in Fig. 1.

(2) Aiming at insufficient processing of local features, an AAF block is proposed to guide the feature learning of low-level contexts and acquire both global and local contexts.

(3) Facing with microcracks with class imbalance issues, a TA block is proposed, which contains spatial attention, channel attention, and pixel attention, to suppress the background and useless information.

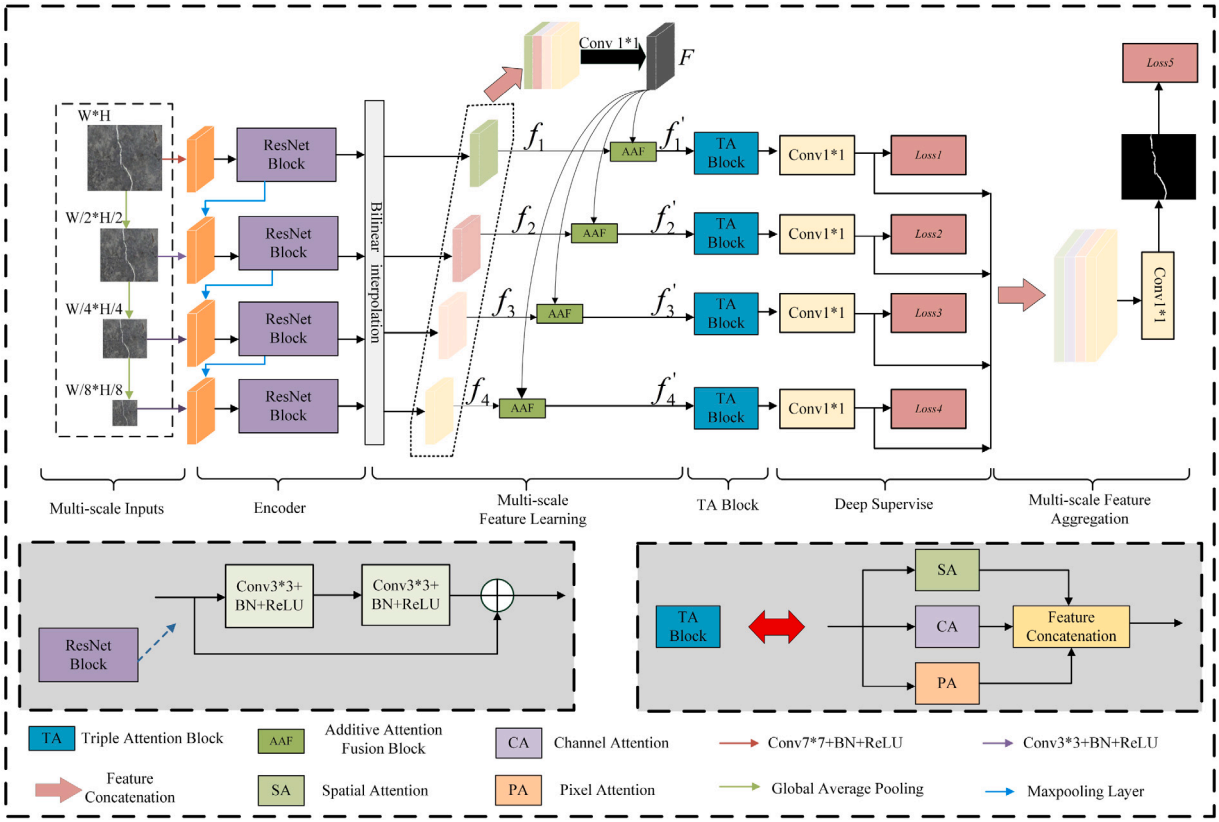

Fig. 2. Network framework of the proposed MST-Net.

(4) To support the deep crack segmentation network to acquire richer feature information, the multiscale input strategy is also introduced. In addition, to enhance the multiscale defect detection ability, a multiscale feature aggregation unit is proposed to make use of multiscale contexts from different receptive fields.

The specific structure of the remaining parts is shown below. Section 2 describes the specific framework of the proposed network and elaborates the details of each proposed network block. Section 3 presents the details of the experimental datasets used and the experimental setup, and Section 4 analyzes and discusses the experimental results. The conclusions of this paper and future directions for further research are presented in Section 5.

2. Methodology

This section describes the specific framework of the proposed MST-Net, the details of each part of the proposed network framework, and the loss function with the deep supervision mechanism.

2.1. Backbone network

The proposed MST-Net is shown in Fig. 2. It starts with a 7 × 7 convolution layer followed by a batch normalization (BN) layer and a rectified linear unit (ReLU) to acquire a large receptive field.

Residual Network: The residual network [26] is a typical deep network and has been widely adopted in different models as the baseline network; it has excellent feature representation performance in different tasks, such as image classification, object detection, image segmentation, etc. Because of the success of residual networks in computer vision, the ResNet18 network is used as the baseline network to build the backbone network. Specifically, the second, third, fourth and fifth residual blocks of the baseline ResNet18 network are adopted for multiscale feature representation.

As shown in Fig. 2, the residual network includes four identical residual blocks, each of which has a similar network structure separately by two 3 × 3 convolution layers followed by a BN layer, a ReLU function, a residual connection, and a max-pooling layer.

Multiscale Input Unit: To support the deep crack segmentation network to acquire richer feature information, a multiscale input strategy is embedded in the backbone network to detect rich feature information. Meanwhile, it can significantly reduce the effect of the information loss issue generated by pooling operations. Here, the multiscale input images from top to bottom can be obtained by averaging pooling layers, which are processed by 3 × 3 convolution layers accompanied by a BN layer and a ReLU function, through which the multiscale feature maps are generated. These feature maps are concatenated with the feature maps from the previous network layer to act as the input of the current network layer.

Multiscale Feature Learning (MFL): To ensure detection precision on multiscale defects fused with global and local feature information, effective multiscale feature aggregation has demonstrated remarkable performance optimization [27–29]. Inspired by this mechanism, feature maps with different resolutions are obtained from the baseline network, which are processed by bilinear interpolation to obtain feature maps with the same feature channels and dimensions f_1, f_2, f_3 and f_4. On this basis, these feature maps at each scale are fused by feature concatenation, which contains local feature information and global context information. The fused feature maps are used to guide the feature learning of low-level feature maps by the proposed AAF block.

To reduce the computing cost, the 1 × 1 convolution layer followed by a BN layer is adopted to process the fused feature maps for channel adjustment to generate the final fused feature F, which provides additive attention fusion guidance for feature maps at each scale, making the extracted features more discriminative.

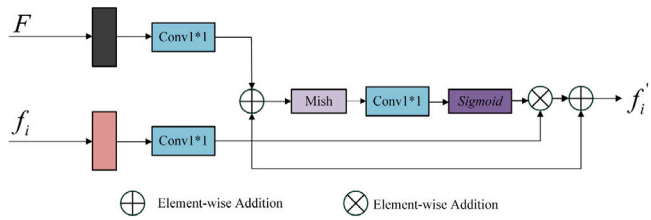

Fig. 3. Network structure of the proposed AAF block.

2.2. AAF block

For rough feature maps generated by a residual network, to enable the extracted features to be more easily identifiable, an AAF block is proposed for additive attention fusion guidance for feature maps at each scale.

As shown in Fig. 3, F stands for the multiscale fusion feature map, which contains not only shallow texture and boundary information but also high-level semantic information, and f_i represents the feature maps after bilinear interpolation. Before feature learning, a 1×1 convolution is applied to two input feature maps to obtain the same channel numbers of F and f_i. These two feature maps are processed by elementwise addition. Furthermore, the attention weight of feature maps can be obtained by a Mish activation function, 1×1 convolution and sigmoid activation function.

$$F_a = \sigma(Conv_{1*1}(Mish(Conv_{1*1}(F) + Conv_{1*1}(f_i)))) \tag{1}$$

where F_a represents the attention weight, $Conv_{1*1}$ represents 1×1 convolution and σ represents the sigmoid activation function.

The obtained attention weight is multiplied by f_i for feature learning of the low-level feature map to enhance the feature representation ability. Meanwhile, the residual connection is also introduced into the proposed AAF block, as shown in Eq. (2).

$$f_i' = Conv_{1*1}(f_i) * F_a + f_i \tag{2}$$

where f_i' represents the feature map by feature learning. Combined with the proposed AAF block, enhanced multiscale feature maps are obtained, which can be used for feature decoding to acquire more accurate segmentation results.

2.3. TA block

The output feature maps generated by the proposed MFL unit include shallow and deep context information from different scales, but simple elementwise addition or channelwise feature concatenation cannot effectively aggregate multiscale feature maps because they not only generate too many parameters but also cause redundancy of feature information [30]. Therefore, to achieve effective feature representation and suppress the background interference in crack images, a TA block is proposed to detect the attention features to better focus on the crack areas. Fig. 4 shows the proposed TA block, which is composed of spatial attention, channel attention and pixel attention.

Fig. 4 shows that the output feature maps of the proposed MFL unit are used as the input of the proposed TA block, and the proposed TA block contains three parallel network branches to focus on key channel information, spatial information and pixel information. The above attention feature maps are fused together by elementwise addition for attention feature fusion to generate final attention feature maps.

Channel Attention: To better focus on key feature channels, a channel attention block is proposed in this paper to construct interdependencies between channels. Here, the channel weights are obtained by a global average pooling layer, a full connection layer followed by a ReLU function and a full connection layer followed by a sigmoid activation function, which is used to multiply the input feature maps f_i' to acquire the channel attention feature maps, as shown in Eq. (3).

$$\mathcal{M}_c^i = \alpha(FC(GAP(f_i'))) \otimes f_i'(i = 1, 2, 3, 4) \tag{3}$$

where M_c^i is the generated channel attention feature map, GAP represents the global average pooling layer, FC represents the full connection layer that tiles the image information into vectors, α represents the sigmoid activation function and $\otimes$ represents elementwise multiplication.

Through the channel attention block, different spatial positions in the feature channel share the same weight, and the interdependencies between channels can be acquired to suppress useless feature channels.

Spatial Attention: To express the information of crack areas more effectively, a spatial attention block is proposed to obtain the spatial attention feature maps to increase the weights of crack regions, as shown in Eq. (4).

$$\mathcal{M}_s^i = \alpha(GAP(f_i')) \otimes f_i'(i = 1, 2, 3, 4) \tag{4}$$

where M_s^i represents the generated spatial attention feature map.

Here, the global average pooling layer enables the feature maps to be compressed into a single channel, and the spatial attention weight can be characterized effectively through a 3×3 convolution with a sigmoid activation function. Combined with the elementwise multiplication between the input feature maps f_i' and spatial attention weight, the spatial attention feature maps are generated to highlight the defect regions.

Pixel Attention: For the pixel attention block, the spatial weight of each channel can be obtained by a 1×1 convolution layer followed by a sigmoid activation function to rescale activations to [0, 1], as shown in Eq. (5). The final pixel attention feature maps are obtained by elementwise multiplication with the input feature maps f_i'.

$$\mathcal{M}_p^i = \alpha(Conv_{1\times1}(f_i')) \otimes f_i'(i = 1, 2, 3, 4) \tag{5}$$

where M_p^i represents the generated pixel attention feature map. Through the pixel attention block, the importance of each pixel in the feature map is emphasized, which further highlights the detailed feature information.

Since each attention mechanism pays attention to a different perspective in the pavement crack segmentation task, the above three attention blocks are fused together to form the proposed TA block, and the specific expression is shown in Eq. (6).

$$\mathcal{M}_i = M_c^i \bigoplus M_s^i \bigoplus M_p^i (i = 1, 2, 3, 4) \tag{6}$$

where M_i represents the aggregated attention feature maps. Combined with the proposed TA block, the spatial attention, channel attention, and pixel attention information are integrated effectively to suppress background interference and provide effective support for accurate pavement crack segmentation with class imbalance issues.

2.4. Deep supervision

The binary cross-entropy (BCE) loss function has shown good performance in various classification tasks, which represents the gap between the true value and the predicted value. Based on the excellent performance of the BCE loss function, it is set as the loss function of the proposed MSTA-Net to guide the network training, as shown in Eq. (8).

$$\mathcal{L}oss_i(p, t) = -\sum_{i=1}^{H*W} t_i \cdot \log p_i + (1 - t_i) \cdot \log(1 - t_i) \tag{7}$$

where p and t represent the predicted value and the true value, respectively.

To enhance the segmentation precision of the proposed MST-Net network and accelerate the convergence speed of the segmentation

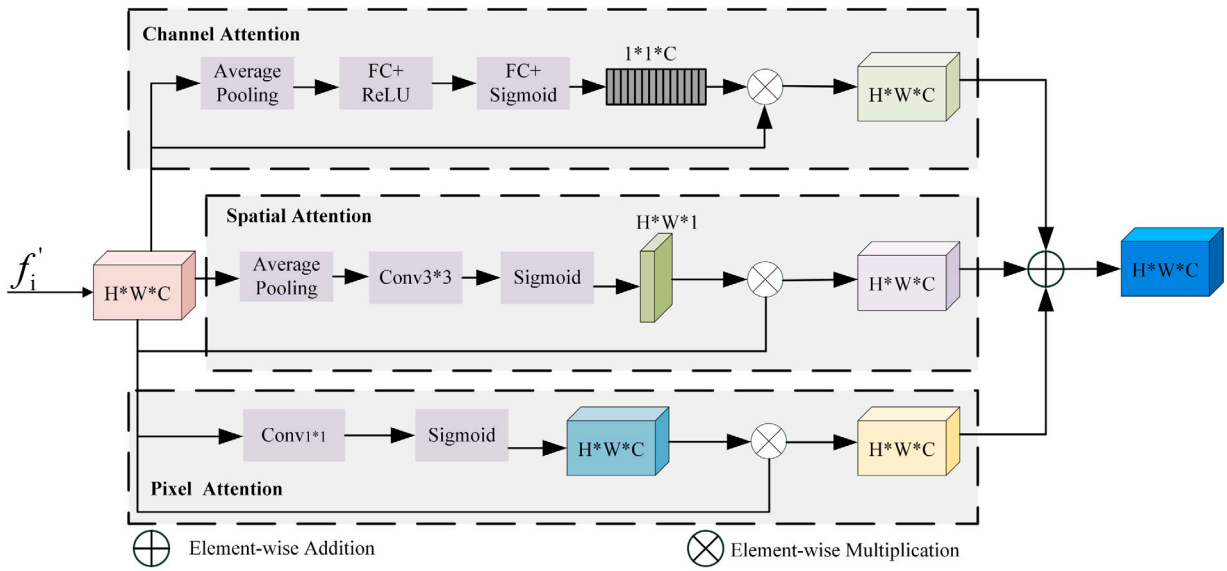

Fig. 4. Network framework of the proposed TA block.

network, a deep supervision mechanism is also adopted for each output layer separately.

For the deep supervision mechanism, the 1 × 1 convolution with softmax function is performed on the four branches to generate four output defect masks. The BCE loss function is applied to the four network branches and final output network branch. The special expression of the total loss function is shown in Eq. (8).

$$\mathcal{L}oss = Loss_1 + Loss_2 + Loss_3 + Loss_4 + Loss_5 \tag{8}$$

The proposed deep supervision mechanism can not only alleviate the problem of too many parameters but also effectively speed up the convergence of the proposed crack segmentation model.

3. Experimental dataset and setup

In this section, the experimental datasets and evaluation metrics used for the performance test and evaluation are first introduced in detail, and the detailed model parameters for the network training are also provided.

3.1. Dataset

For the purpose of demonstrating the superiority of the proposed MSTA-Net on pixelwise crack segmentation tasks, three typical public datasets on crack segmentation, including the DeepCrack537 dataset, YCD dataset and CFD dataset, are used in this paper for experimental analysis and comparison. Fig. 5 shows sample crack images from the above three crack datasets.

As shown in Fig. 5, the above sample images not only contain minor cracks, which occupy the small-scale pixels compared with backgrounds, which cause serious class imbalance issues but also have complex background interference problems. Therefore, the above crack datasets are conducive to the performance evaluation and validation of the proposed MST-Net network. The specifications of these three experimental datasets are shown in Table 1.

DeepCrack537

The DeepCrack537[31] dataset is a common crack dataset collected by Liuet al.. All the crack images in the dataset have corresponding pixel-level cracks, and the image size for all crack images is 544 × 384 pixels. For the segmentation experiments, with a division ratio of 8:2, the training dataset includes 422 images, and the test dataset includes 115 images. The image size is changed to 512 × 512 to test and evaluate the model performance.

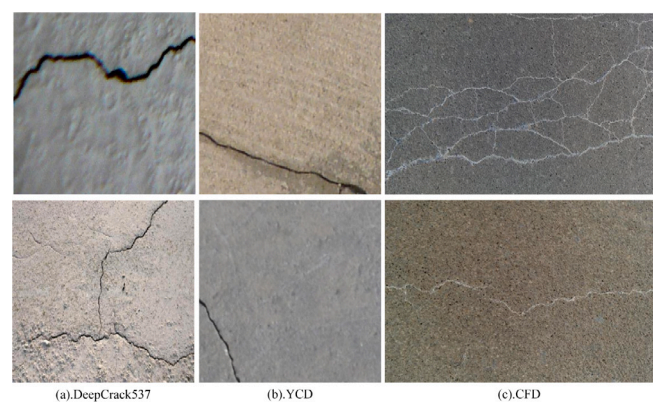

Fig. 5. Sample images in the three public datasets.

Table 1
Specifications of different crack datasets.

Datasets	Image resolution	Training set	Test set
DeepCrack537	512 × 512	422	115
YCD	512 × 512	622	154
CFD	320 × 480	100	18

YCD

The YCD [32] dataset was built by Yang et al., which contained 776 crack images. The image sizes of different images vary. Here, the image size is adjusted to a fixed image size of 512 × 512 pixels. With a division ratio of 8:2, 622 images are used as the training dataset, and 154 images are used as the test dataset.

CFD

The CFD dataset [33] is a pixel-level labeled road crack dataset proposed by Shiet al.. In this dataset, due to the influence of shooting conditions, there is some uneven lighting and image noise, and it also includes some relatively thin cracks, which can be used for performance evaluation. The image size of all the crack images is 480 × 320 pixels. Following a division ratio of 8:2, 100 crack images are adopted for the training dataset, and 18 are adopted for the test dataset.

Combined with the above crack datasets, it can be seen that all the crack datasets are divided into training datasets and test datasets with

Table 2
Experimental parameter setting.

Index	Parameter	Value
1	Learning rate	0.001
2	Batch size	2
3	Optimizer	RMSProp
4	Training rounds	300

a division ratio of 8:2. Combined with the training datasets, to better serve the model training of different deep segmentation models, data enhancement is applied to the training sets to enlarge the crack image datasets, including image transposition and image normalization.

3.2. Evaluation metrics

The crack segmentation network aims to binarize the defect and the background from crack images. Here, to evaluate the superiority of the proposed segmentation network, four evaluation metrics commonly used in segmentation networks are adopted for performance measurement, including Precision, Recall, Dice score, and mIoU value. The detailed mathematical definitions of Precision and Recall are shown in Eq. (9).

$$\begin{cases} Recall = \dfrac{TP}{TP+FN} \\ Precision = \dfrac{TP}{TP+FP} \end{cases} \tag{9}$$

where TP represents true positives, FP represents false positives, TN represents true negatives and FN represents false negatives.

The Dice score evaluates the segmentation performance by assessing the similarity between the predicted values and true values, and the mIoU value evaluates the degree of overlap between the predicted values and true values. The detailed mathematical expressions of the Dice score and mIoU value are shown in Eq. (10).

$$\begin{cases} Dice = \dfrac{2\left|P \bigcap T\right|}{|P|+|T|}, \\ mIoU = \dfrac{\left|P \bigcup T\right|}{\left|P \bigcap T\right|}, \end{cases} \tag{10}$$

where P represents the predicted value and T represents the true value. These two metrics are not easily influenced by class imbalance issues, and they are more accurate metrics for quantitative analysis in segmentation tasks.

3.3. Parameter setting

The proposed MST-Net network is built based on the Baidu PaddlePaddle framework, and all the segmentation experiments are performed on an NVIDIA Tesla V100 GPU with 16 GB memory to speed up the convergence and optimization speed of the segmentation model.

For the training process of the proposed MST-Net network, the learning rate is set to 0.001, and the optimizer is RMSProp. Based on the memory size of the GPU card, the batch size is set to 2, and the total number of training rounds is set to 300. The BCE loss function is used as the loss function, and a deep supervision mechanism is adopted to speed up the model convergence. The detailed parameter setting information is shown in Table 2.

4. Experimental results and discussion

To further prove the superiority of the proposed crack segmentation network, this section combines DeepCrack537, YCD, and CFD datasets to conduct ablation experiments and comparison experiments, which are evaluated by four evaluation metrics to provide precise quantitative analysis. Meanwhile, the visualization results are also presented to provide qualitative analysis on the segmentation performance of the proposed MST-Net network and other advanced segmentation networks.

Table 3
Ablation experiments on the DeepCrack537 dataset.

Method	Precision	Recall	*Dice*	*mIoU*
Baseline	89.3%	87.6%	77.2%	81.1%
Baseline+Multi-scale Inputs	90.8%	89.5%	84.1%	86.1%
Baseline+AFF Block	91.8%	91.8%	85.8%	87.0%
Baseline+TA Block	90.5%	88.9%	86.8%	88.1%
Proposed	**94.9%**	**95.8%**	**90.1%**	**91.1%**

Table 4
Ablation experiments on the YCD dataset.

Method	Precision	Recall	*Dice*	*mIoU*
Baseline	83.0%	76.0%	64.3%	71.5%
Baseline+Multi-scale Inputs	84.9%	78.4%	68.4%	73.1%
Baseline+AFF Block	83.2%	77.1%	68.6%	73.2%
Baseline+TA Block	84.1%	79.1%	67.9%	74.3%
Proposed	**89.4%**	**84.8%**	**75.5%**	**78.7%**

Table 5
Ablation experiments on the CFD dataset.

Method	Precision	Recall	*Dice*	*mIoU*
Baseline	79.3%	76.1%	64.9%	70.9%
Baseline+Multi-scale Inputs	83.9%	80.7%	67.6%	74.3%
Baseline+AFF Block	84.8%	80.6%	67.5%	73.3%
Baseline+TA Block	83.9%	81.9%	67.0%	74.0%
Proposed	**88.7%**	**84.3%**	**73.3%**	**78.5%**

4.1. Ablation results

To verify the effective performance of each network block in the proposed MST-Net network, ablation experiments are conducted on the DeepCrack537, YCD, and CFD datasets. To ensure segmentation precision on the crack images, some network blocks are proposed for performance improvement, including the proposed AFF block, TA block and a multiscale input strategy. The baseline network is the proposed MST-Net network without the above three network blocks. Based on this setting, different network configurations are built for the ablation experiments to analyze the effective performance of the proposed AFF block, TA block and multiscale input strategy, respectively, as follows.

(1) Baseline,
(2) Baseline + Multiscale Input,
(3) Baseline + AFF Block,
(4) Baseline + TA Block,
(5) Proposed MST-Net.

In particular, for the above network configurations, the deep supervision mechanism is adopted for the model training, and the detailed analysis of the effective performance of the deep supervision mechanism will be analyzed in the following subsection.

Four evaluation indicators, including Precision, Recall, Dice score and mIoU value, are used to assess the superior performance of each network block. Combined with the DeepCrack537, YCD, and CFD crack datasets, special ablation experiments on different crack datasets are shown in Tables 3–5.

As shown in Table 3, on the DeepCrack537 dataset, there is a significant improvement in the network performance when different network blocks are added into the baseline network separately, and the proposed AFF block and TA block contribute the most to the enhancement of the whole segmentation performance.

With the above ablation experiment results on the YCD dataset in Table 4, the Dice score improves by 4.1% after adding the multiscale input strategy, but the other three metrics do not improve much because the YCD dataset contains different defects and the texture is relatively complex. Meanwhile, the global and local feature information of the crack images cannot be better extracted with only one multiscale output strategy. Furthermore, by adding the TA block to the baseline

Table 6
Ablation experiment of deep supervision mechanism on three datasets.

Dataset	Method	Precision	Recall	*Dice*	*mIoU*
DeepCrack537	With	94.9%	95.8%	90.1%	91.1%
	Without	92.4%	93.2%	88.6%	89.6%
YCD	With	89.4%	84.8%	75.5%	78.7%
	Without	85.6%	82.4%	70.9%	74.5%
CFD	With	88.7%	84.3%	73.3%	78.5%
	Without	86.7%	78.7%	68.0%	75.4%

network, the segmentation network can better extract the space, channel, and pixel information, which is helpful for crack detection with class imbalance issues. Compared with the other two network blocks, except for a slightly lower *Dice* score, it has the largest contribution to segmentation accuracy in general.

For the ablation experiment results on the CFD dataset in Table 5, it can be seen that with the inclusion of multiscale input, AFF block and TA block, all proposed network blocks have an active impact on the proposed segmentation network. Among the above three network blocks, the proposed AFF block and TA block contribute the most to the improvement of the whole segmentation performance.

To summarize, the ablation experiment results of the proposed MST-Net on the DeepCrack537, YCD, and CFD datasets have proven the rationality and effectiveness of each network block proposed in the paper, while the best experimental results are obtained when these three blocks are combined.

4.2. Effect of loss function

To further explore the impact of the deep supervision mechanism on the segmentation performance, ablation experiments are also conducted on three public crack datasets to verify the effective performance of the deep supervision mechanism.

With four evaluation indicators, including precision, recall, Dice score and mIoU value, and three crack datasets, including DeepCrack537, YCD, and CFD datasets, Table 3 shows the ablation experiment results of the deep supervision mechanism on the proposed MST-Net. It should be noted that *with* indicates that the output results of the four sides are also supervised, and *without* indicates that no deep supervision mechanism is included in the proposed MST-Net.

Combined with the ablation experiment results in Table 6, when the deep supervision mechanism is added into the segmentation network, the metrics (Precision and Recall) improve the most on the DeepCrack537 dataset with increases of 2.5% and 2.6%, respectively. Meanwhile, the Dice score metric improves by up to 4.6% in the YCD dataset, and the Recall metric improves by up to 5.6% in the CFD dataset. In addition, other evaluation metrics on these three public crack datasets have also been improved to varying degrees, which further illustrates the effectiveness of the deep supervision mechanism in this network.

4.3. Comparative experiments

To prove the superior segmentation performance of the proposed MST-Net network on pixel-level crack detection in this paper, some advanced segmentation models are adopted in this paper for performance comparison.

Here, based on four evaluation metrics and three public crack datasets, including DeepCrack537, YCD, and CFD datasets, this subsection shows the comparative experiments on three crack datasets.

Comparative Experiments on DeepCrack537: In recent years, many research works have been published on pixelwise crack segmentation, which have been verified on the DeepCrack537 dataset. Here, to verify the superior performance of the proposed MST-Net network on DeepCrack537, some currently popular mainstream segmentation

Table 7
Comparative results for different segmentation methods on Deepcrack537 dataset.

Method	Precision	Recall	*Dice*	*mIoU*
HED [36]	80.8%	77.2%	79.0%	81.7%
DeepLabv3+[34]	80.4%	81.2%	80.8%	83.2%
U-Net [34]	84.9%	93.1%	77.7%	81.3%
Attention U-Net [35]	92.1%	85.2%	77.5%	80.8%
U-Net3+[37]	95.6%	86.5%	83.0%	83.2%
PSPNet [38]	86.9%	95.2%	81.5%	84.1%
FPHBN [39]	84.0%	80.5%	82.2%	84.1%
HACNet-D [21]	92.6%	80.9%	86.4%	87.4%
DeepCrack [31]	91.9%	78.3%	84.6%	86.0%
DeepCrackT [40]	90.2%	77.5%	83.4%	84.7%
SINet [41]	90.9%	83.1%	86.8%	87.8%
SwinT [44]	91.0%	68.7%	78.3%	81.8%
TransUnet [45]	84.3%	78.7%	81.4%	84.3%
CrackW-Net [23]	**97.7%**	65.7%	75.1%	79.1%
ECDFFNet [42]	90.3%	84.3%	87.2%	88.4%
TCDNet [43]	94.3%	82.7%	88.3%	89.2%
Proposed	94.9%	**95.8%**	**90.1%**	**91.1%**

networks carried out in the DeepCrack537 dataset are used as the comparison models, including CNN-based models (U-Net [34], Attention U-Net [35], HED [36], DeepLabv3+[34], U-Net3+[37], PSPNet [38], FPHBN [39], HACNet-D [21], DeepCrack [31], DeepCrackT [40], SINet [41], CrackW-Net [23], ECDFFNet [42] and TCDNet [43]) and transformer-based models (SwinT [44] and TransUnet [45]). Here, four typical evaluation metrics, including Precision, Recall, Dice score and mIoU value in semantic segmentation, are introduced for performance evaluation. The special experimental results are shown in Table 7, where the best experimental results are indicated by bold font and the second best results are underlined.

Table 7 shows that the proposed MSTA-Net network in this paper has acquired excellent performance compared with some currently popular segmentation networks on the DeepCrack537 dataset, with a precision of 94.9%, recall of 95.8%, Dice score of 90.1%, and mIOU value of 91.1%. Although the precision of the proposed MST-Net network is suboptimal among all the comparison models, it is only 2.8% lower than the best result of CrackW-Net, and the other three metrics far exceed those of the CrackW-Net network.

In addition to Precision and Recall, Dice score and mIOU value are two key evaluation metrics for performance optimization. Recall is used to measure the proportion of correctly predicted crack pixels. The Dice score and mIOU value evaluate the segmentation performance by the similarity between the predicted and true values. The performance of the above three metrics are optimal, with 0.6% improvement of Recall, 1.8% improvement of Dice score and 1.9% improvement of mIOU value than the suboptimal results among all the comparison models, thus further improving the segmentation accuracy effectively and detecting the road crack defects accurately, which further indicate the superiority of the proposed MST-Net network on crack detection and is conducive to road maintenance and repair.

Comparative Experiments on YCD: To further verify the superiority and generalization of the proposed crack segmentation network in this paper, we also conduct an experimental investigation on another public crack dataset (YCD dataset). Here, some currently popular mainstream segmentation networks carried out on the YCD dataset are set as the comparison models. Table 8 shows the comparative results for different segmentation methods on the YCD dataset.

Table 8 shows a comparison test between the proposed MST-Net network and some currently advanced segmentation models on the YCD dataset. The performance of the proposed MST-Net network is superior in all the comparison result categories. Among the four metrics, precision, recall, Dice score, and mIoU value on the proposed MST-Net network are 89.4%, 84.8%, 75.5% and 78.7%, respectively, which are the best results among all comparison models.

Meanwhile, compared with the suboptimal results among all the comparison results, the Precision, Recall, Dice score, and mIoU value

Table 8
Comparative results for different segmentation methods on YCD dataset.

Method	Precision	Recall	*Dice*	*mIoU*
DeepCrack [31]	80.4%	55.2%	65.5%	72.5%
DeepCrackT [40]	78.9%	59.1%	67.6%	73.7%
U-Net [34]	84.7%	73.5%	60.0%	66.9%
Attention U-Net [35]	83.2%	83.3%	68.1%	74.3%
U-Net3+[37]	84.1%	79.1%	67.9%	70.7%
PSPNet [38]	77.2%	72.4%	54.0%	63.7%
HACNet-D [21]	79.6%	62.5%	70.0%	75.2%
Proposed	**89.4%**	**84.8%**	**75.5%**	**78.7%**

Table 9
Comparative results for different segmentation methods on CFD dataset.

Method	Precision	Recall	*Dice*	*mIoU*
HED [36]	55.7%	69.3%	61.7%	44.7%
SegNet [46]	70.6%	74.6%	72.5%	44.7%
CrackSegNet [47]	72.5%	74.7%	73.6%	44.7%
DeepCrack [31]	65.5%	76.0%	70.4%	76.6%
DeepCrackT [40]	67.8%	73.1%	70.4%	76.7%
FFEDN [18]	66.5%	73.9%	70.1%	54.0%
U-Net [34]	82.2%	80.2%	62.8%	72.4%
Attention U-Net [35]	77.4%	**84.6%**	63.0%	71.0%
PSPNet [38]	71.4%	77.9%	49.0%	65.6%
HACNet-D [21]	66.7%	76.0%	71.0%	77.0%
W-segnet [48]	78.4%	80.6%	**79.4%**	70.2%
Proposed	**88.7%**	84.3%	73.3%	**78.5%**

improve by 4.7%, 1.5%, 5.5% and 3.5%, respectively. In addition, as seen from Table 8, although our proposed network is similar to Attention U-Net in terms of Recall, it far exceeds Attention U-Net in the other three metrics, thus further demonstrating the superiority of the proposed MST-Net network on the YCD dataset.

Comparative Experiments on CFD: Similar to the above crack datasets (DeepCrack537 and YCD dataset), the comparison experiments are also conducted on the CFD dataset. The quantitative experimental results of different segmentation models on the CFD dataset are expressed in Table 9.

As shown in Table 9, the proposed MST-Net network acquired 88.7% precision, 84.3% recall, 73.3% Dice score, and 78.5% mIoU value. Compared with other segmentation methods, although the proposed method acquires a lower Dice score than that of CrackSegNet and W-segnet, each metric of our proposed network achieves the optimal results on Precision and mIOU value. For the Recall, it is suboptimal result among all the comparison segmentation models. Specifically, Recall is only 0.3% lower than the optimal result.

Combined with the above comparison experiment results on DeepCrack537, YCD, and CFD datasets, the proposed MST-Net network has obtained a remarkable segmentation performance compared with superior mainstream segmentation models, which demonstrates the superiority and generalization ability of the proposed MST-Net network on pixel-level crack detection.

4.4. Visualization experiments

In addition to the quantitative analysis of the comparative experiments in the three public crack datasets, a qualitative analysis is also conducted for better and direct performance analysis. Similar to the quantitative analysis, some advanced mainstream segmentation networks are also selected as the comparison models for qualitative analysis, including U-Net, Attention U-Net, U-Net3+, and PSPNet.

Based on the above three crack datasets (DeepCrack537, YCD, and CFD datasets), some different crack images from test datasets are adopted for experimental objects for qualitative analysis, and the specific visualization results of the three crack datasets are shown in Fig. 6. In particular, to better observe and compare the visualization results, unideal segmentation areas of prediction masks are marked with a red box.

As shown in Fig. 6, from left to right, each horizontal row represents raw images, ground truth, segmentation masks by the proposed MST-Net network, segmentation masks by U-Net, segmentation masks by Attention U-Net, segmentation masks by PSPNet and segmentation masks by U-Net3+.

For the DeepCrack537 dataset, the proposed MST-Net network can segment cracks more accurately, which are closest to the ground truth, and could serve the maintenance and overhauling of roads. However, some of the other segmentation models have unideal segmentation areas. For example, Attention U-Net has an undersegmentation issue, and UNet3+ has an oversegmentation issue. Meanwhile, the above comparison models also generate varying degrees of segmentation noise.

Meanwhile, it can be seen that the YCD dataset has acquired similar experimental results as the DeepCrack537 dataset. The attention U-Net also has an undersegmentation issue, and UNet3+ has an oversegmentation issue on the YCD dataset. In addition, the segmentation results of the U-Net network produce a large amount of image noise, and the detection results of PSPNet for cracks are intermittent, which is not ideal.

Similarly, for the CFD dataset, the proposed MST-Net network acquired more accurate segmentation masks than other segmentation models, which are closest to the ground truth. For the above comparison models, the above segmentation masks exhibit undersegmentation issues. Meanwhile, varying degrees of segmentation noise are generated during the segmentation masks. Among all the comparison models, PSPNet has the poorest segmentation performance due to the most obvious intermittent cracks.

Based on the above quantitative analysis and qualitative analysis, the superiority of the proposed MST-Net network in this paper on pixel-level crack detection is proven.

4.5. Effect of proposed TA block

To focus on feature information in key regions and suppress useless information, combined with spatial attention, channel attention and pixel attention, this paper proposes a TA block to better characterize crack features and improve the whole segmentation accuracy. Here, to better demonstrate the effectiveness of the proposed TA block visually, heatmaps are introduced for performance analysis. Combined with the experimental dataset, DeepCrack537, Fig. 7 shows the heatmaps of different crack images. As shown in Fig. 7 below, from left to right, each row represents the raw crack images, ground truth, heatmaps generated by the proposed MSTA-Net without the TA block, and heatmaps generated by the proposed MST-Net.

As seen from the presented heatmaps in Fig. 7, the visualization results of the heatmaps for the proposed MST-Net network are closer to the ground truth. Compared with the proposed MST-Net without the TA block, with the introduction of the proposed TA block, the segmentation network acquired more accurate shapes about different cracks, which focused more effectively on the cracked regions due to the attention mechanism. In addition, the heatmaps generated by the proposed MST-Net network contain less image noise. Therefore, the above visualization of the results of heatmaps effectively illustrates the key role of the proposed TA block in this paper on performance improvement.

4.6. Complexity analysis

In addition to the precision evaluation of the proposed MST-Net on the pixelwise pavement crack detection using the four key evaluation metrics, including Precision, Recall, Dice score and mIOU value, to verify its superiority of the proposed MST-Net, the complexity of the segmentation model is also a key criterion to measure the goodness of segmentation models. Here, three key indicators of complexity analysis, model parameters, inference time and FLOPs, are introduced for performance analysis. In addition, some typical segmentation networks

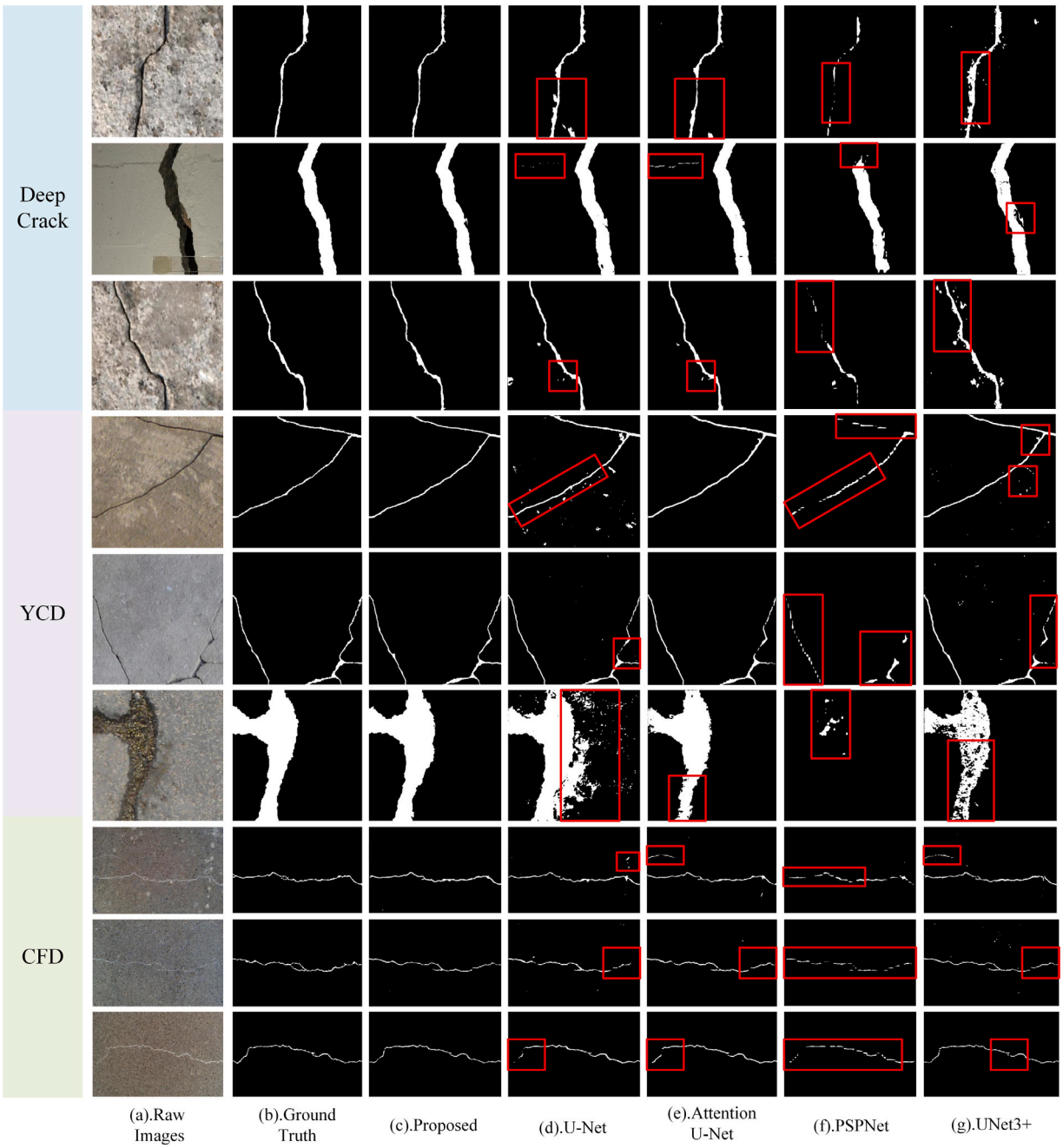

Fig. 6. Visualization results of different crack images.

Table 10
Complexity comparison of different segmentation models.

Model	Parameters (M)	Inference time (s)	FLOPs (T)
U-Net [34]	17.27	0.033	0.16
Attention U-Net [35]	17.62	0.037	0.16
U-Net3+[37]	39.60	0.118	0.93
PSPNet [38]	46.67	0.055	0.18
Proposed	12.97	0.062	0.12

are also set as comparison experiments. Table 10 shows the complexity comparison results of different segmentation models.

As shown in Table 10, it can be seen that the proposed MST-Net has the smallest model parameters and FLOPs among all comparison models. Specifically, the model parameters of U-Net3+ and PSPNet are 3 times and 3.5 times those of the proposed MST-Net network, respectively. For the inference time, the proposed MST-Net is slightly larger than the U-Net model. Although the inference time of the proposed MST-Net is not optimal, the segmentation precision is far better than that of the U-Net model. In summary, the above complexity analysis of different segmentation models shows that the proposed MST-Net has certain advantages in segmentation precision and model efficiency compared with mainstream segmentation models.

4.7. Analysis of failure results

Although the proposed MST-Net network has achieved good performance in the detection of pavement crack defects compared to the currently popular segmentation networks, it still has some issues due to the complexity of the background, poor contrast, weak texture, the

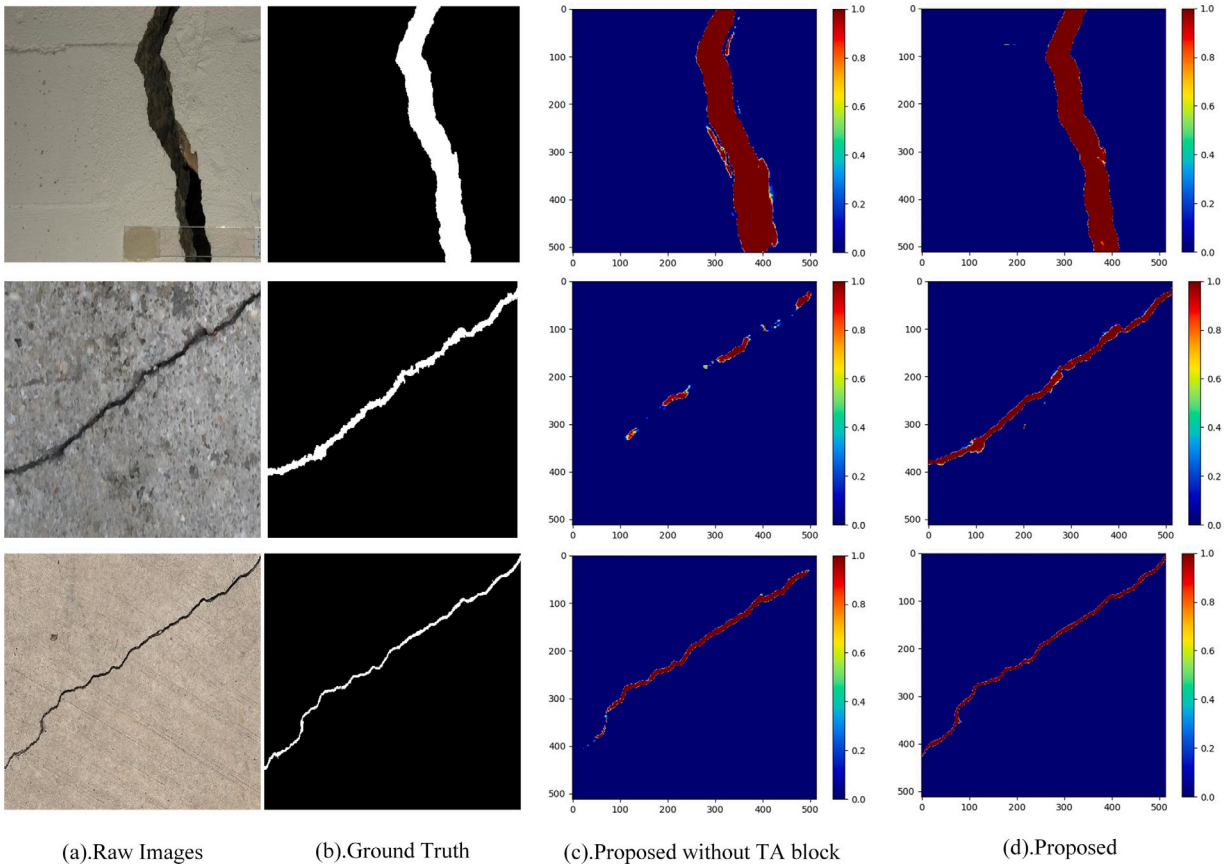

Fig. 7. Visualization results of heatmaps on the proposed TA block.

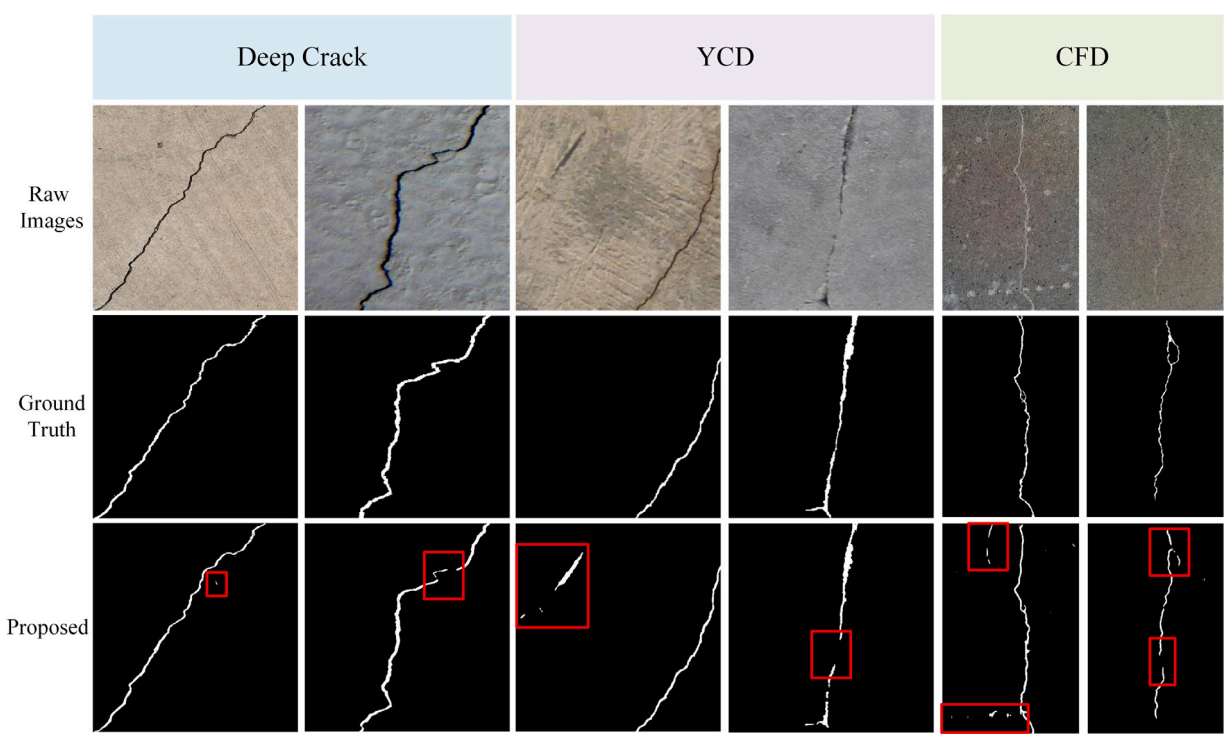

Fig. 8. Failure cases of crack images.

existence of some minor defects, etc. Here, some undesirable results in the three public crack datasets are also presented for qualitative analysis. The visualization results of the failure cases are shown in Fig. 8. To facilitate analysis and observation, the red boxes are used

to mark the unideal areas in the segmentation results of different crack images.

As shown in Fig. 8, some unideal areas are generated due to the similar texture between the pavement defects and backgrounds, and some background pixels are identified as cracks. Meanwhile, because the contrast between some pavement defects and the background is relatively low, the poor texture will also cause undersegmentation. Therefore, complex backgrounds bring a great challenge for high-precision microcrack segmentation. Although the existence of unideal areas will affect the segmentation performance, it will not affect road maintenance and repair.

5. Conclusion

In this paper, an end-to-end deep crack segmentation network, named MST-Net, is proposed for pixelwise pavement defect detection. Combined with multiple crack datasets, the experimental results of quantitative analysis and qualitative analysis show that the proposed MST-Net network exhibits promising segmentation performance on pixelwise pavement defect detection and can meet pavement defect detection demands. The contributions of this paper are shown below.

(1) An automatic and accurate deep segmentation network, named MST-Net, is proposed for pixelwise crack detection.

(2) An AFF block is proposed to provide effective guidance for feature learning at different scales to capture global and local contexts.

(3) A TA block is proposed to focus on spatial attention, channel attention information, and pixel attention information to suppress the background and useless information.

(4) The multiscale input strategy is proposed to support the deep crack segmentation network to acquire richer feature information. In addition, the deep supervision mechanism is introduced to improve the convergence speed of the segmentation network and alleviate the problem of excessive parameter numbers.

Although the proposed MST-Net network achieves good performance in pavement crack segmentation, it still has certain limitations. First, for the analysis of failed results as mentioned above, some samples may have oversegmentation issues as well as missegmentation issues due to the texture similarity of the cracks and background, affecting the whole segmentation performance. Second, the detection speed, while meeting the requirements of crack detection, is still lower than those of some typical segmentation networks. In future research, the network structure of MST-net will continue to be improved to enhance the segmentation accuracy and robustness of crack detection while increasing the detection speed to suit a wide range of high-precision and real-time situations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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